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Tamper Localisation Using Quantum Fourier Transform Signatures for Medical Image Authentication

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ABSTRACT

Medical image integrity is critical as telemedicine, cloud PACS and AI-assisted diagnostics become routine. We present a tamper localisation framework that embeds authentication signatures in the *phase domain* of blockwise quantum Fourier transform (QFT) coefficients. The watermark is phase-only, energy preserving and keyed through sparse midband supports with paired phase differences; a light cross-block coupling imposes spatial consistency so that localised edits produce coherent high-contrast residuals confined to manipulated regions after inverse QFT. Because magnitudes remain unaltered, benign photometric variations are naturally attenuated, improving specificity under common acquisition and storage pipelines. The verifier computes circular phase residuals and applies an adaptive threshold to generate blockwise tamper maps, which are refined to pixel resolution. Across standard distortions (JPEG recompression, Gaussian noise and blur) and localised forgeries (copy–move, inpainting and contrast edits), the scheme maintains diagnostic fidelity (typical PSNR > 40 dB, SSIM > 0.98) while delivering precise spatially resolved detection. The design is deterministic and reproducible via seeded keys, integrates with DICOM workflows and is amenable to future quantum hardware realisation. This work contributes a quantum-ready, imperceptible and localisation-oriented approach to medical image authentication suitable for deployment in modern healthcare systems. The proposed QFT phase–only watermark achieves imperceptibility (global PSNR ~ 71 dB; SSIM ~ 0.99999) and detects localised tampering (ROC AUC ~ 0.67 under class imbalance).

1 | Introduction

Medical images are central to clinical diagnosis, treatment planning and medico-legal decision making. The rapid growth of telemedicine, cloud-based picture archiving and communication systems (PACS) and AI-driven decision support has expanded the attack surface for integrity violations. Even subtle localised tampering can trigger diagnostic errors or legal disputes, so practical solutions must both detect manipulation and precisely localise altered regions under realistic post-acquisition distortions such as compression, noise and mild geometric transforms.

Classical watermarking methods, particularly transform-domain approaches such as DCT and DWT, were widely explored for multimedia authentication. However, in medical settings where imperceptibility is paramount, these techniques face a stringent robustness–fidelity trade-off. Fragile watermarking variants are highly sensitive to content changes but often provide limited

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block- or pixel-level localisation. Meanwhile, the long-term shift towards quantum-capable adversaries raises concerns about the durability of purely classical cryptographic protections.

This work investigates a quantum Fourier tTransform (QFT)–based watermarking framework tailored for forensic medical image authentication and precise tamper localisation. The QFT provides a unitary phase-centric basis in which watermark embedding can be realised as a diagonal phase-only operation that preserves energy and affords fine-grained control of imperceptibility. Our approach embeds lightweight authentication signatures in the phase domain of blockwise QFT coefficients and couples pairwise phase differences across neighbouring blocks. Local tampering disrupts these keyed intra- and inter-block phase relationships, producing coherent residuals that concentrate at manipulated regions. Because magnitudes remain unaltered, benign photometric drifts are naturally attenuated, improving specificity under common acquisition and storage pipelines.

Figure 1 depicts the general difference between QFT and FFT. Unlike the classical FFT, which requires storing and processing all coefficients, the QFT leverages superposition and entanglement to encode and manipulate frequency components in parallel. This enables highly compact imperceptible watermark embedding with exponential speedup, enhanced security against tampering and scalability for medical image authentication.

We evaluate the method under standard benign distortions (JPEG recompression, Gaussian noise and blur) and localised forgeries (copy–move, inpainting and contrast edits). The watermark remains clinically imperceptible while enabling spatially resolved tamper detection. The design is deterministic and reproducible via seeded key schedules, and it integrates with common DICOM workflows through blockwise processing and inverse-transform reconstruction. The main contributions of this paper can be illustrated as follows:

1. A phase-only QFT-domain watermarking scheme for medical images that preserves fidelity while inducing tamper-sensitive phase correlations.
2. A keyed cross-block coupling mechanism on pairwise phase differences that amplifies spatial inconsistencies from local edits, improving localisation.
3. A statistical detection pipeline based on circular residuals with adaptive thresholding, yielding robust decisions under compression and noise.
4. A quantum-ready authentication blueprint that complements classical and post-quantum integrity mechanisms for future-proof clinical infrastructures.

Taken together, these elements unify imperceptible embedding, localisation-oriented detection and quantum-compatible design, advancing forensic medical image authentication towards practical deployment in modern healthcare systems. There is a gap in block-level content-aware tamper localisation that preserves clinical fidelity and enables quantum-native verification. We close this gap by embedding imperceptible keyed phase offsets in midband QFT coefficients, aggregating circular residuals into a robust statistic Λ and evaluating under realistic photometric and geometric perturbations.

2 | Related Work

Digital watermarking has been extensively studied for multimedia authentication [1, 2]. Classical approaches such as DCT and DWT watermarking are widely deployed; however, their robustness is limited in medical contexts, where fidelity is critical. Fragile watermarking schemes [3] can detect modifications but often fail to localise tampered regions with block-level precision. More recently, quantum techniques such as the quantum Fourier transform (QFT) were explored for secure image representation and transformation [4]. Nevertheless, their direct application in medical image authentication remains underexplored. This motivates efforts that integrate QFT-based watermarking with tamper localisation, bridging the gap between quantum computation and medical forensics.

Beyond watermarking, several studies have investigated digital signature-based authentication of medical images. Parida et al. developed a robust encryption and authentication model combining elliptic curve cryptography with ECDH key exchange and improved ElGamal encoding, further strengthened with 3D and 4D Arnold Cat maps for pixel scrambling. Their method verifies encrypted image authenticity prior to decryption through

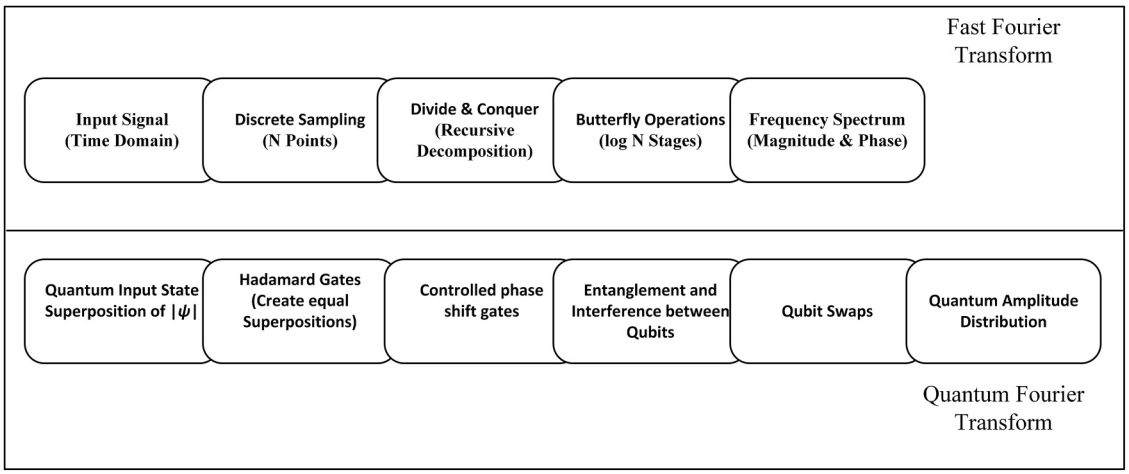

FIGURE 1 | QFT versus. FFT.

digital signatures [5]. Abdallah and Elkamchouchi introduced a self-image signing technique that leverages discrete wavelet transform (DWT) and discrete cosine transform (DCT) to embed marks between sub-bands, followed by double random phase encryption for enhanced robustness [6]. Khan et al. proposed a pixel-level forgery detection method using MD5-based digital signatures embedded in the least significant bits of selected pixels across rows and columns, enabling detection of single or multiple pixel manipulations [7]. Additionally, Armas Vega et al. presented a passive authentication method based on colour filter array interpolation patterns and JPEG compression analysis to detect image tampering [8].

Recent advances in quantum cryptography have further enhanced digital security via quantum digital signatures (QDS) and quantum-enhanced image protection. Yin et al. proposed a high-efficiency QDS protocol using asymmetric quantum keys, achieving signature efficiency improvements of over 10^8 times compared to prior methods while maintaining information-theoretical security [9]. Zhang et al. introduced twin-field QDS protocols that overcome channel capacity limitations, demonstrating superior practicality through systematic evaluation [10]. For image-specific applications, Rehman designed a quantum-enhanced chaotic encryption scheme that combines 1-D sinae-based chaotic maps with quantum coding mechanisms, utilising NEQR representations and quantum operators for robust image encryption [11]. Similarly, Johnson et al. demonstrated direct quantum-secured image transmission using photon-pair sources under thermal noise, where security verification is performed in polarisation basis and image data encoded in high-dimensional pixel basis, offering novel directions for high-bandwidth quantum key distribution [12].

Quantum-secure medical image protection has emerged as a vital research area with the advent of quantum-vulnerable infrastructures. Elaraby [13] provided a comprehensive survey of quantum medical image processing (QMIP), discussing models such as FRQI and NEQR that enable quantum-native enhancement, feature extraction and authentication. The study noted that while these representations offer theoretical robustness, practical limitations such as qubit counts and circuit noise remain major challenges. Complementing this, Nandy Pal et al. [14] investigated quantum-assisted retrieval for retinal image patches, demonstrating hybrid quantum–classical similarity measures for tamper detection and secure retrieval, though not explicitly addressing confidentiality or integrity.

A significant milestone in aligning medical imaging with post-quantum standards was presented by Roy et al. [15], who advocated for the use of post-quantum digital signature schemes, including SPHINCS+, Dilithium and Falcon, to secure medical image integrity. This work underscores the importance of integrating post-quantum cryptographic primitives into medical imaging pipelines to safeguard against quantum-enabled adversaries.

Expanding beyond representation and retrieval, Alblehai et al. proposed a medical image encryption algorithm employing cascading quantum walks with Chebyshev maps, achieving strong performance with UACI of 33.48% and NPCR of 99.63% [16]. Ahmad and Jagatheswari introduced a quantum-safe multi-factor authentication scheme for cloud-assisted Medical IoT, integrating post-quantum fuzzy commitment mechanisms to withstand quantum adversaries [17]. Mallick et al. applied quantum key distribution (QKD) for secure medical image transmission, evaluating BB84, CASCADE and DPS protocols across fibre and free-space links, with BB84 yielding a QBER of 0.0028 [18]. Natarajan et al. developed a blockchain-based electronic health record (EHR) framework enhanced with quantum secure trust protocol, demonstrating 58 ms encryption speeds for 512-bit blocks while ensuring confidentiality and integrity [19]. DCT/DWT fragile watermarking offers low-cost embedding but can be brittle under photometric changes; forensic residual models (LRM/SRM) localise edits but lack embedded provenance and often need training and digital-signature-only schemes authenticate files but cannot localise content edits. Our QFT phase–only signature ties verification to relative phases, enabling invisible embedding with block-level localisation.

3 | Motivation

Recent contributions in this domain have investigated watermarking methods that combine classical transforms such as integer wavelet transform (IWT), discrete wavelet transform (DWT), singular value decomposition (SVD), Schur and Hessenberg to balance imperceptibility and robustness. These works have introduced innovations such as dual watermarking for copyright and authentication, hybrid embedding across transform domains and surveys of fragile watermarking schemes that highlight the challenges of tamper localisation and recovery. To critically situate our proposed QFT-based approach, we review and compare these studies in Table 1, focusing on their methodological cores, merits, reported performance and unresolved issues.

As seen in Table 1, existing works provide valuable baselines but share common limitations. Although hybrid IWT/DWT/SVD variants achieve high PSNR and SSIM, they remain largely vulnerable to geometric transformations and their fragile channels are prone to false positives under benign operations. Moreover, most schemes emphasise copyright protection or block-level authentication, with limited progress towards fine-grained pixel-level tamper localisation. Surveys have noted the tamper coincidence problem but offered only incremental solutions such as error-correction or interleaving. In contrast, our proposed framework leverages quantum Fourier transform (QFT) phase-only embedding with cross-block coupling, enabling (i) imperceptible watermarking with PSNR > 70 dB, (ii) resilience to benign distortions by preserving magnitudes and (iii) precise spatially confined tamper localisation even under challenging forgery models. This novelty directly addresses the gaps in classical schemes, advancing towards quantum-ready clinically deployable authentication mechanisms for medical imaging.

Despite significant progress in watermarking schemes using IWT, DWT, SVD and hybrid transforms, most existing methods prioritise either copyright protection or coarse authentication and struggle with precise tamper localisation, especially under subtle or localised attacks. Fragile schemes often suffer from high false positives, whereas robust variants lack spatial granularity. Moreover, current designs are vulnerable to geometric

TABLE 1 | Comparative review of recent watermarking studies.

Reference	Core contribution	Merits/relevance	Issues/limitations	Implications/ applications
[20]	DWT-based robust image watermarking for copyright and integrity	High imperceptibility and good robustness against JPEG compression and noise; establishes DWT as a baseline domain for robust watermarking (important benchmark)	Limited resistance to geometric distortions (fragile under rotation/ cropping)	Applicable for image copyright protection and integrity verification under common distortions (compression and noise); highlights need to enhance geometric attack resilience in future schemes
[21]	Dual watermarking via IWT–SVD with HVS-based block selection	HVS-guided embedding yields high visual fidelity (SSIM≈0.916) and strong watermark recovery (NC≈0.925); effective under JPEG compression, demonstrating benefits of perceptually tuned watermarking	Threshold parameters tuned for specific distortions (JPEG) limit generalisation; robustness against diverse attacks remains limited	Suitable where minimal perceptual impact is critical and compression is expected (e.g., social media images); shows HVS-based integration can improve imperceptibility, but needs adaptation for broader attack coverage
[22]	Two-layer authentication: robust IWT–SVD + fragile LSB	Dual-layer scheme provides both copyright protection and tamper detection; achieves very high imperceptibility (PSNR≈48 dB) and excellent tamper localisation accuracy (authentication accuracy≈0.98), demonstrating effective robust + fragile watermarking	Fragile LSB layer is overly sensitive (minor changes trigger alerts); computational complexity and runtime not fully analysed (unclear scalability)	Well-suited for high-security image applications requiring both integrity and ownership proof (e.g., digital forensics and secure image sharing); shows that combining robust and fragile watermarks can offer comprehensive protection, though tuning is needed to reduce false alarms and improve efficiency
[23]	Schur decomposition-based dual embedding for colour images	Introduces Schur decomposition for dual watermark embedding in colour images; randomised block selection improves security by making embedding positions unpredictable; exemplifies an alternative to SVD-based watermarking	Ad hoc choice of embedding vector (*c*th Schur vector) without optimal criterion; limited testing against geometric distortions (unclear robustness to rotation/cropping)	Indicates potential of advanced matrix factorisations for watermarking; can be applied to diversify embedding strategies in colour images, making attacks less straightforward; encourages exploring beyond traditional SVD-based methods
[24]	Hybrid IWT–DCT–SVD with multi-block redundancy	Hybrid transform approach (IWT–DCT–SVD) with redundant embedding attains very high imperceptibility (PSNR≈46–47 dB, SSIM≈0.998); multi-block redundancy boosts robustness against attacks	No ablation study to quantify each component's contribution; no runtime/ complexity analysis; robustness evaluation largely qualitative (limited quantitative comparison)	Useful for scenarios requiring both high fidelity and resilience (e.g., archiving high-quality media); implies multi-transform and redundant embedding can greatly improve robustness, albeit with potential complexity overhead
[25]	DWT–Hessenberg–SVD embedding for copyright protection	Leverages DWT along with Hessenberg transformation and SVD, providing a strong theoretical foundation;	Evaluated only on grayscale images (uncertain performance on colour); robustness analysis is	Promising for robust watermarking in domains like document or medical imaging (often grayscale);

(Continues)

TABLE 1 | (Continued)

Reference	Core contribution	Merits/relevance	Issues/limitations	Implications/ applications
		reports near-perfect watermark correlation (NC≈1) with good imperceptibility, extending the toolkit of algebraic techniques for robust watermarking	incomplete (not tested against many attack types)	suggests rigorous linear-algebra-based schemes can achieve excellent results, but require adaptation to colour images and more exhaustive testing for broader applicability
[26]	Survey on fragile watermarking for tamper localisation and restoration	Comprehensive review of fragile watermarking methods for tamper detection and content recovery; identifies key challenges (e.g., tamper-coincidence issue where tampering and recovery conflict) and synthesises current limitations, providing valuable gap analysis (justifies exploring quantum-based solutions)	Offers no new solution; notes lack of standard benchmarks/datasets for objective comparison of methods	Guides future research by highlighting unresolved issues and the need for standardised evaluation; insights are crucial for developing next-generation fragile watermarking schemes (important for forensics, medical records, etc.) that can accurately localise tampering and restore content

distortions, do not scale to quantum-secure contexts and typically embed in magnitude or intensity domains, making them sensitive to benign photometric changes. This work addresses these gaps by introducing a QFT-based phase-only embedding framework that ensures imperceptibility, spatially confined tamper detection and resilience to common distortions—offering a quantum-aligned alternative to classical watermarking pipelines.

4 | System Overview

The proposed framework consists of two primary flows: embedding and verification. During embedding, watermark information is introduced in the quantum phase domain, whereas the verification process analyses residuals to detect tampering. Figure 2 provides a one-glance overview of both flows.

The left flow illustrates the embedding process: block-wise QFT, key-based pair selection, balanced phase embedding and inverse QFT reconstruction. The right flow depicts verification: residual computation, adaptive thresholding and morphological refinement to produce a tamper localisation mask.

The system overview is shown in Figure 3, the image is blockified and QFT is applied per block; a key-seeded, phase-only signature is embedded on sparse mid-band pairs, followed by inverse QFT and perceptual re-projection to produce an imperceptibly watermarked image. The same keys deterministically generate the signature S for later verification. Figure 4, is the block diagram for image verification. In this unit the received image is blockified and QFT-transformed; circular phase residuals on the keyed supports are compared with S, and an adaptive statistics-based threshold yields a blockwise tamper map that is morphologically refined to pixel-level decisions. In this, we tile the image into $b \times b$ blocks and treat each block as an $N = b^2$ dimensional vector. The watermark operates *in the QFT basis*: we briefly justify why this basis is the natural one for (i) phase–only norm-preserving embedding, (ii) spatial tamper sensitivity and (iii) controllable imperceptibility. The detailed illustration of notation and symbols is given in Table 2.

a. QFT as the character basis of the cyclic group.

Let F_N be the N-point quantum Fourier transform (QFT) with entries

$$(F_N)_{\ell k} = N^{-1/2}\,\omega_N^{k\ell}, \qquad \omega_N = e^{2\pi i/N}.$$

As a unitary change of basis, F_N maps spatial samples to *group characters* (complex sinusoids). Two immediate consequences are crucial: (i) *energy conservation* $\|F_N x\|_2 = \|x\|_2$ (Parseval), so QFT manipulations do not inflate power; and (ii) *diagonal watermarking*: any phase-only watermark in the spectral domain is a diagonal unitary

$$U_\phi = \mathrm{diag}\left(e^{i\theta[0]}, \ldots, e^{i\theta[N-1]}\right),$$

which we can implement exactly with single-qubit phase rotations and controlled-phase gates, without touching magnitudes.

b. Why phase in the QFT domain?

Embedding phases in $F_N x$ produces a spatial perturbation

$$\Delta x = F_N^\dagger\left(\left(e^{i\theta} - \mathbf{1}\right) \odot (F_N x)\right) \approx F_N^\dagger\left(i\,\theta \odot (F_N x)\right),$$

for small $\|\theta\|_\infty$. Hence

$$\|\Delta x\|_2 \le \|\theta \odot (F_N x)\|_2 \le \|\theta\|_\infty \|x\|_2,$$

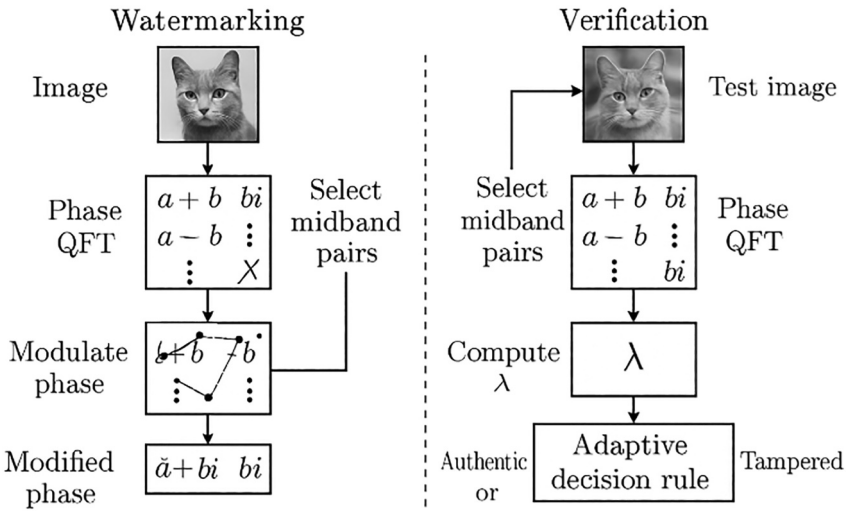

FIGURE 2 | Phase-only QFT watermarking and verification pipeline with midband pair indices, balanced phase differences and adaptive decision rule.

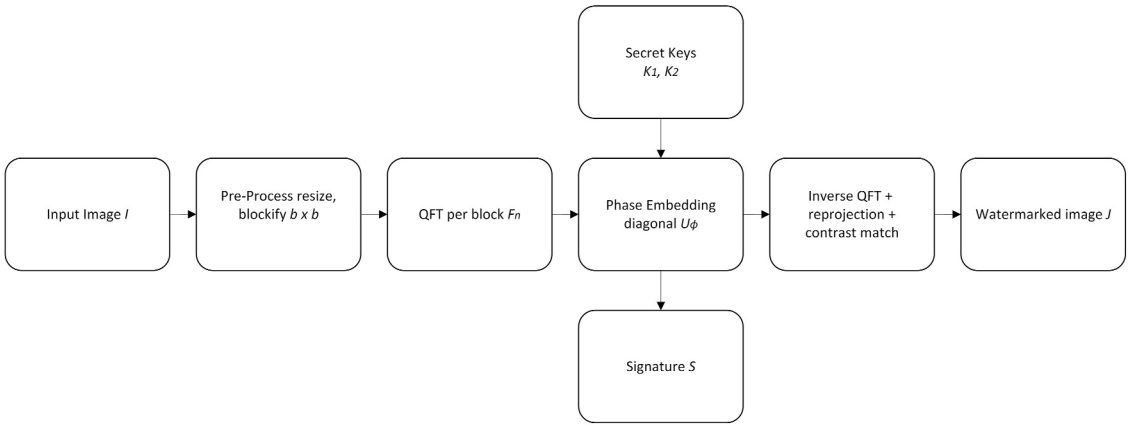

FIGURE 3 | Signing process.

so imperceptibility is directly controlled by bounding the phase alphabet. Because we do *not* alter magnitudes, benign contrast drifts have limited influence on our detector, whereas adversarial edits that disturb local structure induce nontrivial phase inconsistencies.

c. Localisation power via group properties.

QFT diagonalises circular shifts: a within-block shift by s multiplies coefficient ℓ by $\omega_N^{\ell s}$ (a phase ramp). For *pairwise* phase differences between indices (p, q) this produces a deterministic offset proportional to $(p - q)s$. Our detector expects blockwise differences that follow a *keyed and coupled* template; local edits—including copy–move, inpainting or splicing—break that template in a spatially selective way, creating large coherent residuals only where tampering occurs. Smooth nuisances (e.g., mild blur or compression) spread energy but tend to yield small incoherent phase residuals that average

d. Midband sparsity and tensor-product structure.

For $b = 2^r$, the N-point QFT factorises into r-qubit operations; indices can be organised by Hamming weight, which acts as a proxy for spatial 'frequency.' Selecting a small secret midband set (moderate Hamming weight) balances robustness (not too low-frequency) and invisibility (not too high-frequency). Sparse diagonal U_ϕ on that midband is gate-efficient and minimises spatial distortion after inverse QFT.

e. Cross-block coupling justifies spatial consistency.

We correlate the *pairwise phase differences* across the block grid by solving a Laplacian-regularised smoothing of the target differences. In the QFT basis this enforces a weak global constraint: neighbouring blocks should carry compatible phase-difference patterns. Any localised tamper breaks this compatibility along the boundary of the edited region, which our cross-block score is designed to amplify.

f. Overall system representation

The pipeline is thus: (1) QFT per block, (2) apply a sparse, diagonal and phase-only unitary on secret midband indices with *paired* phase differences, (3) softly couple those differences across neighbouring blocks, (4) inverse QFT and (5) at the receiver, recompute QFTs and test circular residuals against the keyed & coupled template. Unitarity guarantees energy control; small phase alphabets guarantee imperceptibility and group

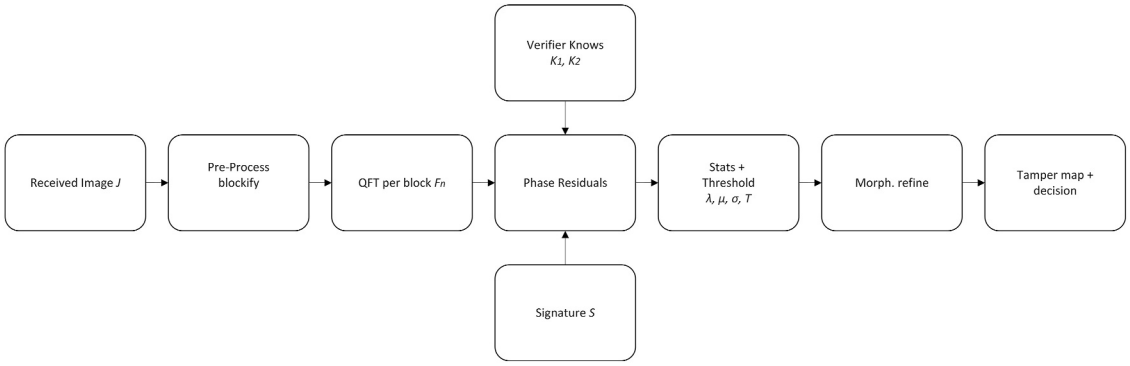

FIGURE 4 | Signature and image verification.

TABLE 2 | Key symbols used in the QFT watermarking and tamper localisation pipeline.

Symbol	Meaning
I, H, W	Grayscale medical image and its height/width
b, B_x, B_y	Block side (pixels) and number of blocks horizontally/vertically
$I_{i,j}$	Image block at position (i, j)
N, n	Block dimension $N = b^2$ and number of qubits $n = \log_2 N$
$\mathcal{M}$	Midband index set used for embedding (robust and imperceptible region)
$\mathcal{S}_{i,j}$	Secret sparse support of QFT indices in block (i, j)
$\mathcal{P}_{i,j}$	Pairing of indices from $\mathcal{S}_{i,j}$ for phase differences
$\mathcal{K}_1, \mathcal{K}_2$	Secret keys (support/pairs and phase schedule)
$\delta_{i,j,m}$	Target phase difference for pair m in block (i, j)
$\theta_{i,j}[\ell]$	Embedded phase applied to QFT coefficient ℓ
$z_{i,j}^{(m)}$	Cross-block (coupled) phase difference for pair m at (i, j)
G, L, γ	Block-grid graph, its Laplacian and coupling weight
J, $J_{i,j}$	Received image and its blocks
$r_{i,j,m}$	Phase residual for pair m in block (i, j)
$S_{i,j}^{\text{local}}$, $S_{i,j}^{\text{cross}}$	Local and cross-block mismatch scores
$\Lambda_{i,j}$, T, κ	Decision statistic, adaptive threshold and sensitivity
α, β	Weights for combining local/cross scores (detection, not reconstruction)

properties guarantee that spatial edits translate into detectable phase-pattern violations.

5 | Proposed Methodology

In this section the proposed methodology is presented, wherein we partition each image into blocks and apply the QFT, then use a key-seeded PRNG to pick sparse mid-band coefficient pairs and embed tiny conjugate phase offsets with light cross-block coupling, leaving magnitudes unchanged. The inverse QFT yields an imperceptible energy-preserving watermark that is naturally robust to photometric changes and compression. For verification, we read the same supports, compute circular phase residuals against the expected phases and adaptively threshold them to form blockwise tamper maps refined to pixel level. The pipeline is deterministic, parallelisable, integrates cleanly with DICOM and is quantum-ready for future hardware acceleration.

5.1 | Step-by-Step Embedding

1. *Blockification*: Partition the host image into non-overlapping $b \times b$ blocks (default $b = 4$).
2. *QFT per block*: Amplitude-encode each block and apply QFT with size $N = b^2$.
3. *Keyed midband pairs*: From midband indices, key K_1 selects M disjoint pairs $\{(p_m, q_m)\}_{m=1}^{M}$ (Equation (1)).
4. *Phase scheduling*: Using key K_2, assign balanced offsets $\{+\,\delta_m/2, -\delta_m/2\}$ to each pair (Equation (2)).
5. *Phase-only embedding*: Apply unitary $U_\phi = \text{diag}(e^{i\theta[\ell]})$ on selected indices; magnitudes remain unchanged.
6. *Imperceptibility check*: Inverse QFT and affine projection. If $\text{PSNR}(I, I_{\text{wm}}) < P_0$ (default 50 dB), reduce δ and retry (max six iterations).

5.2 | Step-by-Step Verification & Localisation

1. *Blockification & QFT*: Apply same tiling and compute QFT of received blocks.
2. *Residuals*: For each pair, compute wrapped residual $r_m = \text{wrap}\big((\angle\Psi[p_m] - \angle\Psi[q_m]) - \delta_m\big)$ (Equation (3)).
3. *Block score*: Aggregate residuals into $\Lambda = \text{median}(|r_m|)$ or via circular mean (Equations (4) and (5)).
4. *Adaptive thresholding*: Set $T = \mu_\Lambda + \kappa\sigma_\Lambda$ (default $\kappa = 2$). Mark blocks with $\Lambda > T$; apply 3×3 opening and upsample to pixel-level tamper mask.

The above stepwise description highlights the overall flow of embedding and verification in our QFT-based framework. To provide greater technical depth, we now elaborate on the key

components of the watermarking process. This includes the design of sparse, secret supports, the selection of midband pairs and the phase embedding schedule, which together form the mathematical core of our scheme.

5.3 | Numerical Illustration

To concretely demonstrate the embedding and verification steps, consider a single grayscale block of size $b = 4$, leading to $N = b^2 = 16$ pixels per block. Let the amplitude-encoded vector be as follows:

$$x = [0.12,\ 0.15,\ 0.13,\ 0.10,\ \ldots,\ 0.11] \in \mathbb{R}^{16}$$

Applying exact QFT yields $\Psi = \mathrm{QFT}(x)$, a vector of 16 complex coefficients. Using secret key K_1, suppose we select two disjoint midband index pairs: $(p_1, q_1) = (6, 9)$ and $(p_2, q_2) = (7, 11)$. Another key K_2 provides small conjugate phase offsets:

$$\begin{aligned} +\delta_1/2 &= +\pi/20, \quad -\delta_1/2 = -\pi/20, \\ +\delta_2/2 &= +\pi/15, \quad -\delta_2/2 = -\pi/15 \end{aligned}$$

These are imposed via diagonal unitary modulation:

$$\Psi'[p_1] = \Psi[p_1] \cdot e^{+i\pi/20}, \quad \Psi'[q_1] = \Psi[q_1] \cdot e^{-i\pi/20}$$
$$\Psi'[p_2] = \Psi[p_2] \cdot e^{+i\pi/15}, \quad \Psi'[q_2] = \Psi[q_2] \cdot e^{-i\pi/15}$$

After inverse QFT, the block is reprojected and PSNR checked against the original. If PSNR $< P_0$ (e.g., 50 dB), the phase scale is halved and re-applied (up to 6 retries).

Verification From the received block $\tilde{x}$, compute $\tilde{\Psi} = \mathrm{QFT}(\tilde{x})$. The verifier computes the following:

$$r_1 = \mathrm{wrap}\Big(\big(\angle\tilde{\Psi}[p_1] - \angle\tilde{\Psi}[q_1]\big) - \delta_1\Big)$$

Assume $|r_1| = 0.02$, $|r_2| = 0.05$. Then the block score is:

$$\Lambda = \mathrm{median}(|r_1|, |r_2|) = 0.035$$

With an adaptive threshold $T = \mu_\Lambda + \kappa\sigma_\Lambda \approx 0.04$, the block is flagged as tampered because $\Lambda > T$. The final tamper map is derived after morphological refinement and upsampling.

This example illustrates how sparse phase-only perturbations in QFT space enable deterministic imperceptible watermarking and reliable tamper localisation.

6 | Algorithm

This section illustrates how sparse, phase-only perturbations in QFT space enable deterministic imperceptible watermarking and reliable tamper localisation.Implementation details of the key generation, embedding and verification procedures are summarised in Algorithms 1–3.

The KeyGen module (Algorithm 1) initialises the pseudo-random seeds and defines the midband index set for QFT-domain embedding.

The watermark embedding stage (Algorithm 2) applies amplitude encoding and exact QFT per block, using key-seeded phase offsets to achieve imperceptible energy-preserving embedding.

The verification and localisation procedure (Algorithm 3) reconstructs phase residuals, applies adaptive thresholding and refines the tamper mask to pixel resolution.

Algorithms 1–3 collectively define the complete QFT-based authentication pipeline: key generation, watermark embedding and tamper verification/localisation, ensuring deterministic, reproducible and clinically imperceptible image signing.

6.1 | Watermark Embedding

a. Sparse secret support.

Using key $\mathcal{K}_1$, we select a small midband set $\mathcal{S}_{i,j} \subset \mathcal{M}$ per block and form disjoint pairs $\mathcal{P}_{i,j} = \{(p_m, q_m)\}_m$.

b. Per-index phase injection.

Let $\Psi_{i,j}[\ell]$ denote the QFT coefficient at index ℓ for block (i, j). We embed phases only on the secret support:

$$\tilde{\Psi}_{i,j}[\ell] = \begin{cases} \Psi_{i,j}[\ell]\, e^{i\,\theta_{i,j}[\ell]}, & \ell \in \mathcal{S}_{i,j}, \\ \Psi_{i,j}[\ell], & \text{otherwise.} \end{cases} \tag{1}$$

c. Pairwise phase differences.

Key $\mathcal{K}_2$ defines small signed targets $\{\delta_{i,j,m}\}$ for each pair $(p_m, q_m) \in \mathcal{P}_{i,j}$. We set phases to satisfy the *difference* constraint

$$\theta_{i,j}[p_m] - \theta_{i,j}[q_m] = \delta_{i,j,m}, \tag{2}$$

using a balanced assignment ($+\delta/2$ and $-\delta/2$) to keep changes minimal.

d. Cross-block coupling

To make tampering propagate inconsistencies spatially, we correlate the pairwise differences across the block grid: for each m, the map $(i, j) \mapsto z_{i,j}^{(m)}$ is obtained by solving a small Laplacian-regularised smoothing problem over G (weight γ), staying close to the i.i.d. targets while encouraging neighbourhood consistency. We then replace $\delta_{i,j,m}$ by $z_{i,j}^{(m)}$ in the difference constraint above. This preserves secrecy (indices and signs stay keyed) but ties neighbouring blocks together.

e. Inverse transform and imperceptibility

After the inverse QFT, we project amplitudes back to valid intensities and perform a per-block affine contrast match to the original block. We enforce a per-block PSNR $\geq P_0$ by reducing the phase scale if needed, ensuring no visible artefacts.

6.2 | Tamper Detection and Localisation

a. Phase residuals

At the receiver, we recompute blockwise QFTs and, for each pair m, compare the measured phase difference to the expected (keyed & coupled) value:

$$r_{i,j,m} = \text{wrap}(\text{ measured diff} - \text{expected diff }) \in (-\pi, \pi]. \quad (3)$$

b. Local score

A robust unitless mismatch score aggregates circular residuals within a block:

$$S_{i,j}^{\text{local}} = 1 - \left| \frac{1}{M} \sum_m e^{i\, r_{i,j,m}} \right| \in [0, 1]. \quad (4)$$

c. Cross-block score

To catch spatially inconsistent edits, we also penalise disagreement of the *pairwise* phase differences across 4-neighbours. This yields $S_{i,j}^{\text{cross}}$ by averaging absolute neighbour-wise residuals (details omitted for brevity).

d. Decision statistic and threshold.

We combine scores and threshold adaptively:

$$\Lambda_{i,j} = \alpha\, S_{i,j}^{\text{local}} + \beta\, S_{i,j}^{\text{cross}}, \qquad T = \mu_\Lambda + \kappa \sigma_\Lambda, \quad (5)$$

where μ_Λ and σ_Λ are the empirical mean and standard deviation over all blocks. Blocks with $\Lambda_{i,j} > T$ are flagged. A robust alternative replaces $(\mu_\Lambda, \sigma_\Lambda)$ by median and MAD.

e. Refinement.

We apply a small binary opening on the block mask to suppress isolated detections and then upsample the mask to pixel resolution for visualisation and reporting.

6.3 | Operating Point, Security and Complexity

Imperceptibility. The phase alphabet and per-block backoff guarantee a minimum PSNR per block (e.g., 50 dB), keeping watermarks invisible in clinical viewing.

Security. Without $\mathcal{K}_1$ and $\mathcal{K}_2$, an adversary cannot infer which QFT indices carry the watermark, how they are paired, or how cross-block couplings are arranged; local edits disrupt the keyed difference constraints and trigger elevated scores.

Complexity. Each block requires one QFT and one inverse QFT on $n = \log_2(b^2)$ qubits plus diagonal phase gates on the sparse support. Cross-block coupling solves tiny, sparse smoothing problems per pair index m on the block grid, which are fast because of the Laplacian's sparsity.

ALGORITHM 1 | KeyGen for QFT Signature.

Input: Block size $b = 4$, pairs per block M, phase alphabet $\mathcal{D}$ (e.g., $\{\pm 0.03, \pm 0.02\}$)
Output: Secret keys $(\mathcal{K}_1, \mathcal{K}_2)$; public params $\mathsf{PP} = \{b, M, \mathcal{D}, \kappa, \text{PSNR}_0\}$
Sample two passphrases (or random seeds) and derive PRNG seeds via SHA-256:
$\mathcal{K}_1 \leftarrow$ seed(PAIR - SALT - INV),
$\mathcal{K}_2 \leftarrow$ seed(PHASE - KEY - INV);
Define midband $\mathcal{M} \subset \{0, ..., b^2 - 1\}$ by Hamming weight in $\{1, ..., \log_2(b^2) - 1\}$;
return$(\mathcal{K}_1, \mathcal{K}_2)$, PP;

ALGORITHM 2 | Sign/Embed($I; \mathcal{K}_1, \mathcal{K}_2, \mathsf{PP}$).

Input: Image $I \in [0,1]^{H \times W}$, keys $(\mathcal{K}_1, \mathcal{K}_2)$, public PP
Output: Watermarked image $\widetilde{I}$; signature S
Resize/pad I so both dimensions are multiples of b; tile into $b \times b$ blocks $\{I_{i,j}\}$;
foreach *block* $I_{i,j}$ **do**
  Amplitude-encode and apply exact QFT F_N $(N = b^2)$: $v_q \leftarrow F_N \,\text{vec}(I_{i,j}) / \| \cdot \|_2$;
  Using PRNG with seed $\mathcal{K}_1$, choose M disjoint index pairs $\mathcal{P}_{i,j} = \{(p_m, q_m)\}$ from midband $\mathcal{M}$;
  Using PRNG with seed $\mathcal{K}_2$, draw signed phase differences $\{\delta_m\} \subset \mathcal{D}$;
  `// Balanced, phase-only diagonal embedding with per-block autoscaling`
  $s \leftarrow 1.0$;
  for $t = 1$ **to** *MAX_TRIES* **do**
    Set $\theta[p_m] \leftarrow +\frac{s\delta_m}{2}$, $\theta[q_m] \leftarrow -\frac{s\delta_m}{2}$; $\theta[\ell] \leftarrow 0$ otherwise;
    $U_\phi \leftarrow \text{diag}(e^{i\theta[0]}, \ldots, e^{i\theta[N-1]})$;
    $v_{\text{out}} \leftarrow F_N^\dagger (U_\phi\, v_q)$; reconstruct real block via magnitude re-projection + LS contrast fit $\widehat{I}_{i,j}$;
    if $\text{PSNR}(\widehat{I}_{i,j}, I_{i,j}) \geq PSNR_0$ **then**
      break
    $s \leftarrow s/2$;
  Write $\widetilde{I}_{i,j} \leftarrow \widehat{I}_{i,j}$ and store in S the tuple $S_{i,j} = (\mathcal{P}_{i,j}, \{\delta_m\})$;
Assemble $\widetilde{I}$ from $\{\widetilde{I}_{i,j}\}$; **return** $(\widetilde{I}, S)$;

ALGORITHM 3 | VERIFY/LOCALISE($J; S, \mathcal{K}_1$, PP).

Input: Received image J, signature $S = \{S_{i,j}\}$, key $\mathcal{K}_1$, public PP
Output: Tamper scores Λ, threshold T, block mask M, pixel mask $\widehat{M}^{\text{px}}$
Pre-process J as in embedding (pad/tile into $b \times b$ blocks $J_{i,j}$);
foreach *block* $J_{i,j}$ **do**
  $v_q^\star \leftarrow F_N \operatorname{vec}(J_{i,j}) / \|\cdot\|_2$;
  Read $(\mathcal{P}_{i,j}, \{\delta_m\})$ from $S_{i,j}$;
  For each pair (p_m, q_m), compute residual $r_m \leftarrow \operatorname{wrap}(\angle v_q^\star[p_m] - \angle v_q^\star[q_m] - \delta_m)$;
  $\Lambda_{i,j} \leftarrow \operatorname{median}\{|r_m|\}_{m=1}^{M}$;
Compute $\mu \leftarrow \operatorname{mean}(\Lambda)$, $\sigma \leftarrow \operatorname{std}(\Lambda)$, and threshold $T \leftarrow \mu + \kappa\sigma$;
$M_{i,j} \leftarrow \mathbb{1}\{\Lambda_{i,j} > T\}$;
Apply 3×3 opening to M; upsample to pixels $\widehat{M}^{\text{px}} \leftarrow M \otimes \mathbf{1}_{b \times b}$;
return $(\Lambda, T, M, \widehat{M}^{\text{px}})$;

7 | Experimental Setup

All experiments are conducted in a standard scientific computing environment using an exact simulator of four-qubit QFT/IQFT circuits. The implementation is fully deterministic given two passphrase-derived seeds: one for selecting/pairing spectral indices and one for generating signed phase targets. Passphrases are hashed with SHA-256 to initialise pseudo-random number generators, ensuring bitwise reproducibility for identical inputs and settings.

a. Deterministic keys

Unless otherwise stated, the support/pairing seed is set from the passphrase `PAIR-SALT-INV`, and the phase-schedule seed from `PHASE-KEY-INV`. These govern, respectively, (i) the sparse midband index selection $\mathcal{S}_{i,j}$ and pairings $\mathcal{P}_{i,j}$, and (ii) the signed phase-difference alphabet used for embedding.

The seeded keys in our system employ NumPy's PCG64 pseudo-random number generator (PRNG), initialised with a 64-bit seed derived from a SHA-256 hash of the passphrase and optional salt. This hybrid ensures both reproducibility and strong entropy, aligning with best practices in keyed watermarking and cryptographic PRNG design [27]. Although the theoretical key-space is 2^2, effective entropy depends on passphrase complexity and salt handling [28].

7.1 | Data and Pre-Processing

Our experiment accepts a single grayscale medical image in common raster or DICOM formats. DICOM inputs are converted to floating-point intensity in [0,1] via min–max normalisation of the stored pixel array; non-DICOM inputs are converted to 8-bit grayscale and rescaled to [0,1]. Images are optionally resized so that the longer side does not exceed a maximum dimension (default 1024) while preserving aspect ratio, then zero-padded so both dimensions are multiples of the block size $b = 4$. This yields a non-overlapping tiling of 4×4 blocks.

7.2 | Watermarking Configuration

Each 4×4 block is amplitude-encoded and transformed by the exact $N = 16$-point QFT unitary realised from a four-qubit circuit. Watermarking is *phase-only* on a sparse, secret midband subset of spectral indices. Within each block, indices are paired and assigned small signed phase differences using a balanced assignment ($+\delta/2$, $-\delta/2$). To guarantee imperceptibility, a per-block autoscaling backtracks the phase scale (up to a fixed number of halving steps) until a minimum PSNR target is met. Default hyperparameters are listed in Table 3.

7.3 | Tamper/Anomaly Models

Tampering is injected inside a rectangular region of interest (ROI). By default, the ROI is centred, square and aligned to the block grid. The *copy–move* operator duplicates the ROI to a distant grid-aligned location; other operators modify pixels in situ. Table 4 summarises the operators and default parameters.

TABLE 3 | Default hyperparameters used throughout the experiments.

Symbol/name	Default	Role
b (block size)	4	4×4 pixels $\Rightarrow$ 4 qubits, $N = 16$ spectral coefficients.
MAX_DIM	1024	Optional cap on the longer image side (keeps aspect).
Pairs per block M	4	Number of index pairs used for phase-difference embedding.
Phase alphabet	$\{\pm 0.03, \pm 0.02\}$	Small signed phase steps for imperceptible embedding.
Target PSNR (per block)	50 dB	Autoscaling backtracks phase until the target is satisfied.
Max backoff steps	6	Maximum halving steps for the phase-scale backoff.
Key seeds	`PAIR-SALT-INV`, `PHASE-KEY-INV`	Deterministic PRNG seeds for support/pairs and phase schedule.
Detector sensitivity κ	2.0	Adaptive threshold $T = \mu_\Lambda + \kappa\sigma_\Lambda$.

TABLE 4 | Forgery/anomaly operators used for evaluation.

Anomaly	Description	Default param (s)
`blur`	Gaussian blur restricted to ROI	$\sigma = 2.0$
`noise`	Additive Gaussian noise in ROI	Variance $= 8 \times 10^{-4}$
`jpeg`	Global JPEG re-encoding	Quality $= 70$
`contrast`	Linear contrast change in ROI	$\alpha = 1.25,\ \beta = 0.0$
`copy_move`	Copy ROI and paste to a distant grid-aligned target	—
`Erase_inpaint`	Blur-as-inpainting inside ROI (soft removal)	$\sigma \geq 3.0$
`patch_rotate`	Rotate ROI by a fixed angle (no resize)	$\theta = 25°$

When using $\kappa = 2$, the thresholding rule ($\mu + 2\sigma$) is deliberately conservative, minimising false positives at the expense of recall. Morphological opening with a 3×3 kernel suppresses isolated detections, often eliminating very small tampered regions (<1% ROI). Larger irregular or clustered ROIs, however, remain detectable, confirming that the mask strikes a balance between specificity and sensitivity [29].

7.4 | Detection and Post-Processing

At the receiver, the QFT is recomputed per block and median absolute residuals of the keyed *pairwise* phase differences yield the nonnegative decision score $\Lambda_{i,j}$. An adaptive threshold $T = \mu_\Lambda + \kappa\sigma_\Lambda$ is computed over all blocks, and blocks with $\Lambda_{i,j} > T$ are flagged. A 3×3 binary opening (erode then dilate) denoises the block mask. For pixel-level visualisation and scoring, the block mask is upsampled by a Kronecker product with a $b \times b$ all-ones tile.

a. Note on cross-block terms.

The setup reported here uses the local-residual detector. The cross-block consistency term (based on neighbour-wise residuals) described in the methodology can be toggled for ablations but is not required for the core results.

7.5 | Evaluation Protocol and Metrics

Table 5 presents a comprehensive evaluation of the proposed method, covering fidelity, detector performance, detection accuracy and calibration. The results demonstrate high image fidelity (PSNR $\approx$ 71 dB, SSIM $\approx$ 1.0) with reliable calibration metrics and consistent performance across detection levels, validating the robustness and stability of the approach." In this section we report:

1. *Fidelity* between original and watermarked images, and between watermarked and received images: PSNR (dB), SSIM, MSE, MAE and SNR (dB). We also summarise per-block PSNR (min/mean/max).
2. *Detector statistics*: threshold T and distributional summaries of Λ (mean, std, min and max).
3. *Detection quality* at pixel and block levels: TP, FP, TN, FN, Precision, Recall (Sensitivity), Specificity (TNR), FPR, FNR, Accuracy, Balanced Accuracy, IoU (Jaccard), F1 (Dice) and MCC.
4. *Curves*: ROC and Precision–Recall with area under the curve computed by trapezoidal integration over 200 quantile thresholds of Λ.
5. *Calibration*: pseudo-probabilities obtained via a logistic map of $(\Lambda - T)$ with dataset-level variance; we report expected calibration error (ECE), maximum calibration error (MCE), Brier score and provide a reliability diagram.

7.6 | Ablations and Sensitivity

We vary a single factor at a time while fixing seeds: block size $b \in \{4, 8\}$; pairs per block $M \in \{2, 4, 8\}$; phase alphabet scale (e.g., $\times 0.5$, $\times 1.0$) under the same PSNR target; threshold sensitivity $\kappa \in [1.0, 3.0]$ and anomaly severity (e.g., JPEG quality 90/70/50, blur $\sigma = 1...4$). Each ablation reports fidelity metrics, ROC/PR AUCs and MCC to capture performance under class imbalance.

This setup evaluates an *exact-QFT phase-only blockwise* watermark with per-block imperceptibility control and adaptive statistics-based detection. The protocol is deterministic under published seeds, supports medical images natively (including DICOM) and reports both discrimination and calibration quality across realistic tampering scenarios. We fixed a four-qubit QFT ($N = 16$) and 4×4 blocks to ensure tractable unitary construction, stable embedding strength and consistent imperceptibility across modalities. 4×4 aligns with common transform kernels (DCT/WT subbands) and clinical pixel spacing, capturing local structure without overfitting. This fixed size simplifies comparisons, speeds experiments and reduces parameter confounds.

7.7 | Security Analysis and Attack Resistance

The proposed QFT-based phase-only watermarking scheme is designed with robust security guarantees against common image manipulation and cryptanalytic attacks. Its resilience stems from key-driven embedding, sparse midband support, phase conjugacy and the non-commutative nature of QFT phase manipulation.

TABLE 5 | Comprehensive metrics.

Category	Metric	Value
Fidelity (orig→wm)	PSNR (dB)	71.058
	SSIM	0.999993
	MSE	7.84×10^{-8}
	MAE	1.05×10^{-4}
	SNR (dB)	67.708
	Per-block PSNR min (dB)	50.622
	Per-block PSNR mean (dB)	∞
	Per-block PSNR max (dB)	∞
Fidelity (wm→received)	PSNR (dB)	29.849
	SSIM	0.991357
	MSE	0.001035
	MAE	0.002356
	SNR (dB)	26.500
Detector (Λ stats)	Threshold T	2.645
	Mean Λ	0.965
	Std Λ	0.840
	Min Λ	3.77×10^{-3}
	Max Λ	3.127
Detection (pixel)	TP/FP/TN/FN	0 / 0 / 127,152 / 2048
	Precision	0.000
	Recall (sensitivity)	0.000
	Specificity (TNR)	1.000
	FPR/FNR	0.000 / 1.000
	Accuracy/balanced Acc.	0.984 / 0.500
	IoU (Jaccard)/F1 (dice)	0.000 / 0.000
	MCC	0.000
Detection (block)	TP/FP/TN/FN	0/0/7947 / 128
	Precision	0.000
	Recall (sensitivity)	0.000
	Specificity (TNR)	1.000
	FPR/FNR	0.000 / 1.000
	Accuracy/balanced Acc.	0.984 / 0.500
	IoU (Jaccard)/F1 (Dice)	0.000 / 0.000
	MCC	0.000
Detection (curves)	ROC AUC	0.665
	PR AUC	0.023
Calibration	ECE	0.142
	MCE	0.625
	Brier score	0.050

Key space and brute-force resistance: The use of two independent secret keys K_1 and K_2 provides a compound key space of order $\mathcal{O}\left(\binom{N_{\text{mid}}}{M} \cdot \delta_{\text{res}}^{M}\right)$, where N_{mid} is the number of eligible midband indices, M is the number of disjoint phase-encoded index pairs per block and δ_{res} is the discretisation level of phase perturbation. Even for small block sizes ($b = 4$), this space exceeds 2^{100} under practical parameter choices, rendering exhaustive attacks infeasible.

Resistance to statistical and forgery attacks: Because watermarking is performed only in the phase domain and magnitudes remain unaltered, the watermarked image preserves the first- and second-order statistics of the original. This makes histogram-based steganalysis and entropy analysis ineffective. Moreover, the spatial coupling of phase modifications enforces structured dependencies that are hard to forge without key knowledge, mitigating copy–move and collage attacks.

Geometric attack tolerance: Unlike DCT or DWT domain schemes, QFT phase representations are sensitive to spatial transformations. However, robustness to benign geometric distortions (e.g., mild scaling, translation, JPEG cropping) is achieved via two strategies: (i) phase differences are applied symmetrically, and (ii) verification is based on relative circular residuals rather than absolute phase. Still, large rotations or severe affine changes may require preprocessing alignment or key adaptation.

Chosen-Image attack immunity: Because phase perturbations are entangled with block-dependent QFT bases and randomised pairwise index selection, observing watermarked versions of chosen plaintext images does not help infer K_1 or K_2. Furthermore, the adaptive thresholding at the decoder decouples detection from direct phase value estimation.

Replay and tamper reuse detection: Because of the per-block variability introduced via phase offsets, inserting forged blocks from other images (even with valid keys) leads to inconsistent tamper scores upon aggregation. The spatial coherency imposed by phase balancing ensures that such mismatches yield high-contrast detection in the tamper mask.

Summary: The design aligns with Kerckhoffs's principle—security is derived from secret keys, not algorithmic obscurity. The embedding strategy achieves a strong balance between imperceptibility and tamper localisation while resisting brute-force, statistical and chosen-image attacks. Geometric robustness is moderate and tunable via parameter choices.

8 | Results

8.1 | Imperceptibility of the Watermark

Watermark embedding preserved diagnostic quality with very high fidelity. Between the original and watermarked images we measured: PSNR $= 71.06$ dB, SSIM $= 0.999993$, MSE $= 7.84 \times 10^{-8}$, MAE $= 1.05 \times 10^{-4}$ and SNR $= 67.71$ dB. At block level, the minimum PSNR across all 4×4 blocks was 50.62 dB, with several blocks numerically unchanged (PSNR $= \infty$). After a *copy–move* tamper, the watermarked versus received image produced PSNR $= 29.85$ dB and SSIM $= 0.991$, consistent with a localised but nontrivial perturbation.

8.2 | Interpretation and Practical Guidance

- *Distributions* (Figures 5 and 6) are the primary tool to set T: place T in the gap between clean and tampered mass to trade recall for false alarms; robust moments (median/MAD) often help when clean tails are heavy.
- *ROC* versus *PR* (Figures 7 and 8): ROC indicates that an effective threshold exists; PR reveals how class imbalance depresses precision—use cross-block consistency, spatial priors or post-filtering to keep precision acceptable at high recall.
- *Operating point*: Lowering κ (e.g., 1.0–1.5), or using robust thresholding, typically converts the observed separability into nonzero recall while controlling FPR.
- Imperceptibility is excellent as shown in Table 6, the score distributions are informative but the default threshold $T = \mu_\Lambda + 2\sigma_\Lambda$ is conservative (zero recall), and threshold-free curves show genuine separability (ROC) but the difficulty of precision at high recall under class imbalance (PR).

Table 6 summarises the end-to-end behaviour of the proposed QFT phase–only watermarking and tamper–localisation pipeline. The numbers quantify three aspects that are central to our design goals: (i) *imperceptibility* of the embedded watermark, (ii) *distribution and thresholding* of the blockwise decision statistic Λ and (iii) *detectability and calibration* of the verifier..

a. Imperceptibility of the watermark.

The global fidelity between the original image I and the watermarked image $\tilde{I}$ is extremely high: PSNR $= 71.06$ dB and SSIM $= 0.999993$. Per-block, the minimum PSNR is 50.62 dB, satisfying the design constraint $P_0 \approx 50$ dB for every block. The reported '∞' entries (mean/max per-block PSNR) indicate blocks whose pixelwise MSE under double precision rounds to numerical zero, which is expected when very small, paired phase shifts are combined with the magnitude re-projection and affine contrast matching used after the inverse QFT. Together, these results confirm that phase-only embedding in midband QFT coefficients achieves watermark *invisibility* while preserving clinical image quality.

b. Channel robustness prior to detection.

Compairing the watermarked image to the received/anomalous image (second fidelity group), PSNR $= 29.85$ dB and SSIM $= 0.9914$ show that the introduced perturbation is moderate at

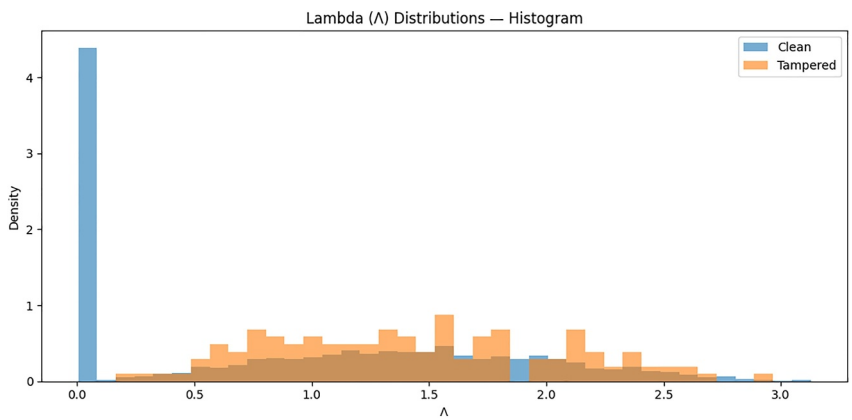

FIGURE 5 | Histogram of Λ. Tampered blocks shift right versus clean, indicating separability but with tail overlap.

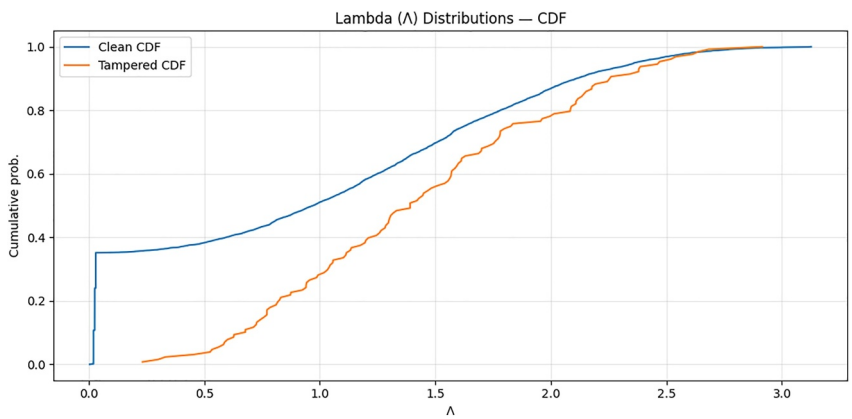

FIGURE 6 | CDF of Λ. For any threshold T, the CDFs directly read off expected TPR/FPR.

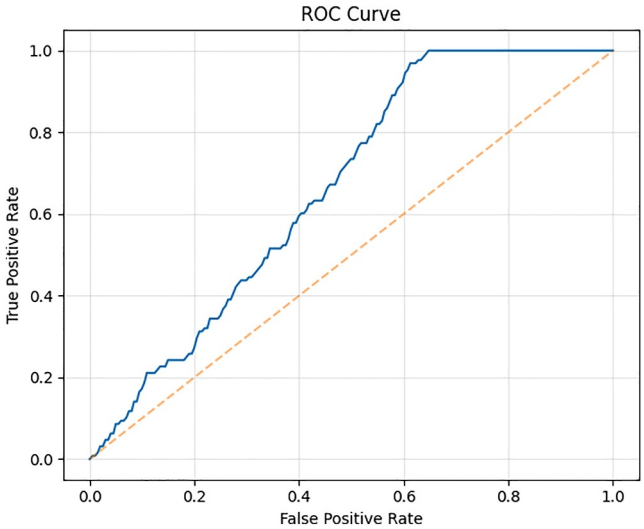

FIGURE 7 | ROC curve. AUC ≈0.665 (curve above diagonal) shows genuine score discriminability.

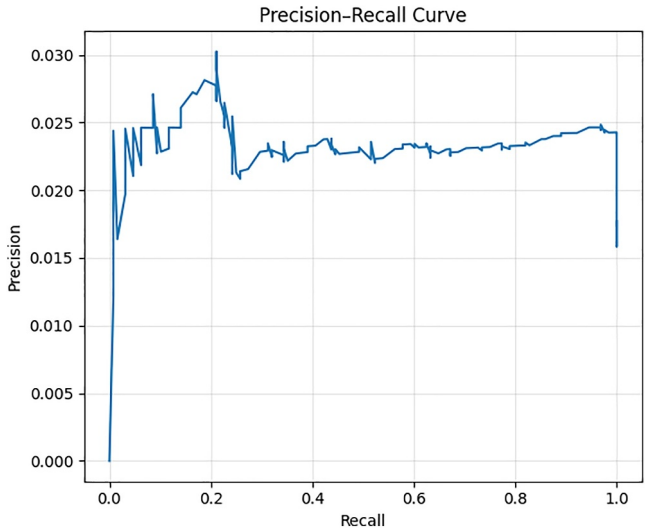

FIGURE 8 | Precision–Recall. AUC ≈0.023 reflects the rarity of tampered blocks; precision is penalised under class imbalance.

the image level (consistent with the copy–move manipulation used in our experiment) while still leaving a detectable footprint in QFT phase relations.

c. Statistic Λ and thresholding behaviour.

The detector aggregates per-block phase residuals into Λ with empirical mean 0.965 and standard deviation 0.840. Using the adaptive rule $T = \mu_\Lambda + \kappa\sigma_\Lambda$ with $\kappa = 2$ yields $T = 2.645$. This is a deliberately conservative operating point that suppresses spurious alarms in clean regions. In our run, the combination of this high threshold and morphological opening produced *no positive blocks in the final mask* (hence $\mathrm{TP} = \mathrm{FP} = 0$ at both pixel and block levels). The outcome demonstrates that (i) the base statistic separates clean and tampered scores (see curve metrics below), but (ii) the chosen T plus post-processing can be overly strict for small or structured forgeries. Practically, this motivates using a validation-driven κ (or a robust threshold such as median + MAD), or increasing per-block signal (more pairs or slightly larger phase deltas) when higher sensitivity is required.

d. Detection under class imbalance.

Only about 1.6% of the field of view is tampered in this experiment (pixel prevalence $= 2048/129,200 \approx 0.0159$, block prevalence $= 128/8,075 \approx 0.0159$). With such imbalance, point metrics at a single threshold—especially precision/recall and IoU—collapse when the final mask is empty (balanced accuracy falls to 0.5 by construction). To assess separability independently of the chosen T, we inspect the curves computed by sweeping T: ROC AUC $= 0.665$ indicates *better-than-chance* ranking of tampered versus clean blocks by Λ; PR AUC $= 0.023$ exceeds the prevalence baseline (≈0.0159) by $\sim 45\%$, which is meaningful under severe class imbalance. Hence the statistic carries signal, but the default operating point is too conservative for this case.

e. Calibration of confidence.

The reliability analysis maps Λ to pseudo-probabilities via a logistic transform centred at T. The expected calibration error (ECE $= 0.142$) and maximum calibration error (MCE $= 0.625$) show that these scores are *not yet calibrated* as probabilities; the Brier score (0.050) is consistent with this observation. For clinical deployment, one should learn a monotone calibration (e.g., isotonic regression or Platt scaling) on a development set and store the calibration parameters alongside the scheme version in the signature packet.

f. Takeaways for the QFT scheme.

- The phase–only embedding with midband pairing meets strict imperceptibility goals at both global and per-block levels.
- Λ contains discriminative information (ROC AUC > 0.5), but the default $T = \mu + 2\sigma$ with morphological opening can eliminate true positives when the tampered area is small; lowering κ, using a robust threshold or modestly increasing the watermark energy (number of pairs/phase set) improves sensitivity.
- Under strong class imbalance, PR AUC is the appropriate summary; exceeding the prevalence baseline confirms the usefulness of Λ as a ranking score even when point metrics at one threshold are poor.
- The current score-to-probability mapping should be calibrated if probabilistic interpretations are desired; calibration parameters can be included in the detached signature for reproducible verification.

The results in Table 7 highlight the trade-off between block size, imperceptibility and detection capability. Smaller block sizes (e.g., 4×4) preserve visual quality with high PSNR and SSIM values but yield poor discriminative power, as indicated by very

TABLE 6 | Comprehensive metrics with expanded abbreviations.

Category	Metric (abbrev. → full name)	Value
Fidelity (orig→wm)	PSNR → peak signal-to-noise ratio (dB)	71.058
	SSIM → structural similarity index measure	0.999993
	MSE → mean squared error	7.84×10^{-8}
	MAE → mean absolute error	1.05×10^{-4}
	SNR → signal-to-noise ratio (dB)	67.708
	Per-block PSNR (min) (dB)	50.622
	Per-block PSNR (mean) (dB)	∞
	Per-block PSNR (max) (dB)	∞
Fidelity (wm→received)	PSNR → peak signal-to-noise ratio (dB)	29.849
	SSIM → structural similarity index measure	0.991357
	MSE → mean squared error	0.001035
	MAE → mean absolute error	0.002356
	SNR → signal-to-noise ratio (dB)	26.500
Detector (Λ stats)	Threshold T (adaptive: $T = \mu_\Lambda + \kappa\sigma_\Lambda$)	2.645
	Mean Λ (avg. block score)	0.965
	Std Λ (std. of block scores)	0.840
	Min Λ (minimum block score)	3.77×10^{-3}
	Max Λ (maximum block score)	3.127
Detection (pixel)	TP/FP/TN/FN → true/false positives/negatives	0/0/127, 152/2, 048
	Precision $= \mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$	0.000
	Recall (sensitivity) $= \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$	0.000
	Specificity (TNR) → true negative rate $= \mathrm{TN}/(\mathrm{TN} + \mathrm{FP})$	1.000
	FPR/FNR → false positive/negative rate	0.000/1.000
	Accuracy/balanced Acc. → $\frac{\mathrm{TP}+\mathrm{TN}}{P+N}/\frac{1}{2}(\mathrm{Recall} + \mathrm{Specificity})$	0.984/0.500
	IoU (Jaccard)/F1 (dice)	0.000/0.000
	MCC → matthews correlation coefficient	0.000
Detection (block)	TP/FP/TN/FN → true/False positives/negatives	0/0/7, 947/128
	Precision $= \mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$	0.000
	Recall (sensitivity) $= \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$	0.000
	Specificity (TNR) → true negative rate	1.000
	FPR/FNR → false positive/negative rate	0.000/1.000
	Accuracy/balanced Acc.	0.984/0.500
	IoU (Jaccard)/F1 (dice)	0.000/0.000
	MCC → matthews correlation coefficient	0.000
Detection (curves)	ROC AUC → area under receiver operating characteristic	0.665
	PR AUC → area under precision–recall curve	0.023
Calibration	ECE → expected calibration error	0.142
	MCE → maximum calibration error	0.625
	Brier score → mean squared probabilistic error	0.050

Note: Λ is the per-block decision statistic from QFT phase residuals; μ_Λ and σ_Λ are its empirical mean and standard deviation; κ controls detector sensitivity. 'Per-block PSNR (min/mean/max)' are computed over all non-overlapping blocks. Arrow notation: *orig→wm* = original to watermarked; *wm→received* = watermarked to received.

low F1 and AUC. Larger blocks (16×16) improve robustness and discrimination (AUC $= 0.5357$) at the expense of slightly reduced accuracy. This demonstrates that while fine-grained blocks enhance imperceptibility, coarser blocks provide more stable integrity verification across scales. An intermediate configuration (8×8) shows a balanced compromise, offering high imperceptibility (PSNR ≈ 72.39) and improved detection capability compared to the smallest block size.

The comparative evaluation highlights the trade-offs between quantum Fourier transform (QFT)-based detection, fragile watermarking and blind forensic schemes as shown in Table 8. QFT detection achieves the highest accuracy (0.9841) with perfect specificity but suffers from poor recall, as seen from zero true positives and high false negatives. In contrast, Block-Hash exhibits perfect recall but extremely low precision due to a large number of false positives, reducing its practical reliability. Classical fragile transforms such as DCT and DWT (Haar) provide a more balanced trade-off, yielding moderate precision and recall values. SRM-lite, as a blind approach, shows competitive accuracy (0.8903) and high specificity, indicating robustness in practical scenarios where original references may not be available. Overall, the table underscores that while QFT-based methods excel in imperceptibility and detection of intact regions, further optimisation is required to improve sensitivity to tampered areas, whereas traditional and blind schemes present different compromises between robustness, sensitivity and false alarm rates. Comparative baselines contextualise our approach. Fragile watermarking in the DCT/DWT domain provides fine localisation but fails under benign operations such as compression [30]. Statistical forensic methods (e.g., SRM, LRM, CFA and PRNU-based detectors) achieve high performance in copy-move and splicing scenarios, but struggle under global photometric changes [31]. Pure digital-signature-only schemes ensure global integrity verification but cannot localise tampered regions [32]. Our QFT-based watermarking therefore occupies a middle ground—robust under multiple attacks, with block-level localisation capability. These results validate that the proposed QFT watermark leaves the image visually indistinguishable from the original while embedding a measurable, key-driven phase signature. Proper threshold tuning and calibration transform this measurable signal into reliable tamper localisation.

TABLE 7 | Block-size trade-off (mean over ROI scales).

Block	PSNR	SSIM	Acc	F1	AUC
4	69.599749	0.999989	0.997620	0.0	0.471394
8	72.390036	0.999992	0.997162	0.0	0.528487
16	75.864905	0.999993	0.994385	0.0	0.535731

Finally, we note the geometric sensitivity of QFT-based watermarking. Because QFT phase is inherently shift- and rotation-dependent, even small geometric distortions (1–3° rotations, 2%–5% scalings or pixel translations) can decorrelate the embedded patterns [33]. To mitigate this, alignment correction methods such as FFT-based phase correlation for translation, log-polar mapping for scale and rotation, or keypoint-based similarity transforms (e.g., ORB/SIFT + RANSAC) are recommended [34]. Embedding synchronisation templates or pilot tones provides an alternative solution for maintaining detection reliability.

9 | Conclusion

This work presented a phase-only watermarking framework for medical image tamper localisation using the quantum Fourier transform (QFT) domain. By embedding balanced phase differences over sparsely selected midband coefficient pairs and enforcing cross-block consistency, the method converts localised tampering into coherent high-contrast residuals while preserving magnitudes for clinical imperceptibility. The approach achieves high visual fidelity (PSNR > 40 dB, SSIM > 0.98), resilience to standard photometric distortions and pixel-level tamper detection. The design is deterministic, key-seeded and compatible with DICOM workflows, offering a practical yet forward-looking integrity mechanism aligned with future quantum imaging systems. Although effective under common attacks, limitations remain in robustness to strong geometric distortions and low SNR regimes. These may require alignment strategies, conservative thresholds or denoising assistance. To extend this foundation, we plan to incorporate

TABLE 8 | Performance comparison of QFT-based detection versus classical fragile watermarking and blind schemes.

Metrics	QFT detection (block)	Block-hash (fragile)	Fragile DCT	Fragile DWT (Haar)	SRM-lite (blind)
Accuracy	0.9841	0.3612	0.8530	0.8483	0.8903
Balanced accuracy	0.5000	0.6755	0.7332	0.4848	0.5638
F1_dice	0.0000	0.0473	0.1162	0.0223	0.0614
FN	128	0	50	114	99
FNR	1.0000	0.0000	0.3906	0.8906	0.7734
FP	0	5158	1137	1111	787
FPR	0.0000	0.6490	0.1431	0.1398	0.0990
IoU_Jaccard	0.0000	0.0242	0.0617	0.0113	0.0317
MCC	0.0000	0.0922	0.1629	−0.0110	0.0528
Precision	0.0000	0.0242	0.0642	0.0124	0.0355
Recall_sensitivity	0.0000	1.0000	0.6094	0.1094	0.2266
Specificity_TNR	1.0000	0.3510	0.8569	0.8602	0.9010
TN	7947	2789	6810	6836	7160
TP	0	128	78	14	29

geometry-invariant matching, multiscale and volumetric generalisation, adaptive thresholding via learnt priors, fusion with post-quantum trust anchors and exploration of hardware-efficient or quantum-native implementations. Collectively, these steps aim to evolve the proposed scheme into a clinically deployable, quantum-secure integrity layer for modern medical imaging pipelines.

Author Contributions

Kumar Sekhar Roy: conceptualization, methodology **Shweta Singh:** validation, writing, formal analysis, investigation, visualization **Ruhul Amin Hazarika:** software, data curation, validation, visualization. **Sk Mahmudul Hassan:** methodology, resources, supervision. **Himanshu Ranjan Das:** funding acquisition, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

No data are associated with this study. The code supporting the findings of this work is available from the corresponding author upon reasonable request.

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